Fiscal Year:	FY 2024	**Task Last Updated:**	FY 01/24/2024
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Project Title:	Reduced-Order Model and Experiments Using the PSI Data from STDCE		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
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Comments:			
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No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
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Task Description:

It is proposed to utilize Physical Sciences Informatics (PSI) data of the Surface Tension Driven Convection Experiment (STDCE) to validate a low-dimensional model which takes interfacial deflections into account. This method will then be compared with a full numerical model that also tracks the interface deformation and flow dynamics of long-wave flows that accompany such experiments. The STDCE data is vital as it will be used via modern imagining software (eg., Track-Mate in Image J) to analyze flow profiles and thermal data. The full numerical model and the simplified low-order method will, upon validation, be employed to design and run a ground experiment whose objective is to use novel means via forced mechanical resonance to quench the long wave surface tension gradient flow.
The STDCE experiments were designed to understand the role of surface tension gradient-driven thermocapillary flow, often encountered in technologically relevant processes such as crystal growth. It was found that long wave steady two-dimensional convective flow gives way to oscillatory flow for certain parameter ranges. These long-wave flows are often detrimental to the relevant technological processes.

The advantage of using a reduced-order method in addition to the full numerical model is that interface evolution may be resolved via a semi-analytical and tractable set of equations for two state variables. This feature provides great insight into the physics of the convection and the root cause of the onset of oscillatory flow from a steady base flow. The nonlinearity of the model facilitates tracking of the interface deformation, which plays a significant role in the onset of oscillatory instability in STDCE. The PSI data will be invaluable as it will provide validation for the model by providing information from image analysis of that data.

The idea behind the proposed experiment is to generate thermal convection in a test-cell that emulates the STDCE and increase the forcing temperature difference until oscillatory flow is triggered. This test cell will be placed on a shaker, currently available in the Principal Investigator's laboratory, whose shaking frequency is set and amplitude is incrementally raised until the oscillatory flow is arrested and even the steady long wave flow is reduced. This is predicted to occur because resonant instabilities act on interfacial deformation and such deformation is characteristic of the STDCE. The critical amplitude to quench the undesired flow can be then compared with theory. The proposed ground-based experiments will identify operating parameter spaces where oscillatory instability may be suppressed. The semi-analytical model will also help identify the root causes for and suppression of oscillatory flow.

In summary, the PSI data is needed to validate the full numerical model and also the low-dimensional model which, in turn, allows us to extract the physical phenomena. The proposed ground experiment will provide additional validation for the model and also give information on suppression of convection via resonant forcing. The results of the study will be of great benefit to materials processing including crystal growth and in-space welding as well as additive manufacturing using directed energy deposition where free surface convection inevitably takes place.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

The research impacts are:
a) Low dimensional models that use length scale separation and which extract dominant physics in problems of fluid instability would be validated by complete models that do not make such approximations. This will allow low dimensional models to explain physical models with great simplifications. b) Obtaining a clear means on thwarting surface tension driven convection by means of resonant forcing.

The Earth Benefits are:

a) Potential technologies for improved heat transfer in micropipe heat exchangers b) Potential improvement for additive manufacturing

Task Progress:

The Surface Tension Driven Convection Experiment (STDCE) is a clear demonstration of flows driven by surface-tension gradients along an interface in the absence of gravity. Such flows are termed Marangoni flows. Their strength depends on the container geometry and the scaled temperature gradient (i.e., the Marangoni number) and other dimensionless groups of thermophysical properties such as the Prandtl number, capillary number, and Biot number. The STDCE configuration is such that a steady "base" flow is immediately set up for all non-zero temperature gradients, however small they may be. But when the temperature gradient becomes large (i.e., when the Marangoni number increases beyond a critical value), flow oscillations commence from the steady base flow state and indeed were seen in a follow-up experiment to STDCE-1.
The microgravity environment of STDCE precludes any buoyancy driven flows, which ordinarily manifest as short wave length patterns. In the absence of gravity, flows are typically long wave in character. This means that such flows may be well-described by reduced-order methods that use different length scales for the vertical and horizontal coordinates. Such models account for inertia and dissipating effects with amazing precision by using a technique called the weighted residual integral boundary layer (WRIBL) method and have the distinct advantage of analytically displaying the essential physics of the flows and the ensuing instabilities via evolution equations that describe interface dynamics. This happens because the reduced-order models ultimately lead to dynamic equations for the surface position and surface temperature as a function of transverse position and time. Reduced order models are applicable when there is a separation in length scales between two competing effects such as thermal/momentum inertia and viscous/thermal diffusion. However, they are not applicable when, say, turbulence and chaos sets in. This, of course, is not the case of the STDCE experiment. The main goal of understanding the physics of long wave Marangoni flows and the transition to oscillations is thereby facilitated by the use of reduced order models. Such long wave models have been successfully bench-marked by DNS for a large number of physical problems, most of which are connected to interfacial instability with very diverse physical origins.

We have two general aims in this project. The first is to validate low dimensional models at various levels of length scale ratios with STDC data, as the latter is well suited to such analysis. A second equally important aim is to use low dimensional modeling to predict experimental behavior where we might strive to quench Marangoni convection by enforcing suitable mechanical resonant forcing (i.e, by using Faraday forcing).

In the first year of this project, we have focussed on the second aim. Thus, a major discovery in the current project is that oscillatory mechanical accelerations can cause resonance which, in turn, can quench flow caused by surface tension driven convection.

Bibliography Type:	Description: (Last Updated: 06/17/2026)
Articles in Peer-reviewed Journals	Dinesh B, Livesay J, Ignatius IB, Narayanan R. "Pattern formation in Faraday instability—experimental validation of theoretical models." Philos. Trans. R. Soc. A. 2023 Apr 17;381(2245):20220081. https://doi.org/10.1098/rsta.2022.0081 , Apr-2023